

Tutorial 2 –The Instruments of Trade Policy

ECON 308 – International Economic Relations

Fall 2025

Problem 1 – Tariff and Welfare for a Small Country

Home imports wheat at a world price $P_W = \$200$ per ton. Domestic supply and demand functions are:

$$Q_S = P - 100, \quad Q_D = 700 - P,$$

where Q is measured in tons and P in dollars.

- (a) Compute equilibrium imports under free trade.
- (b) The government imposes a **specific tariff** of \$20 per ton. Determine the new domestic price, domestic quantity supplied, quantity demanded, and imports.
- (c) Calculate tariff revenue.
- (d) Compute the deadweight losses from both production and consumption distortions.
- (e) Determine the net welfare loss.

Problem 2 – Large Country and the Terms of Trade Effect

Home faces a foreign export supply curve:

$$P^* = 100 + 0.5Q^*,$$

where P^* is the foreign export price (in dollars per ton). Home's import demand is:

$$Q_M^D = 400 - 2P,$$

where P is the domestic import price.

- (a) Find the free-trade equilibrium price and quantity of imports.
- (b) Home imposes a tariff of \$20 per ton. Compute the new world price, the new domestic price, and the volume of imports.

- (c) Calculate tariff revenue and the resulting terms-of-trade gain for Home.
- (d) Determine whether welfare rises or falls by comparing the terms-of-trade gain to the deadweight losses.
- (e) Briefly explain why retaliatory policies from trading partners may prevent welfare gains in reality.

Problem 3 – Effective Rate of Protection

A car sells internationally for \$20,000. Its imported engine costs \$8,000 and is subject to a 25% tariff. Finished cars face a 10% tariff.

- (1) Compute the **nominal protection rate** on the finished car.
- (2) Compute the **effective rate of protection (ERP)**:

$$\text{ERP} = \frac{V'_A - V_A}{V_A},$$

where domestic value added is V_A without tariffs and V'_A with tariffs.

- (3) Interpret whether the industry's effective protection is greater or less than the nominal tariff.